Care Access Navigator

Incident Response 1 — Task 1: IR Environment Setup

One lab for all six IR tasks: a Parrot OS VM with a Wazuh agent, a Wazuh manager, Wireshark, Volatility, and macOS log collection

Document ID	CAN-IR1-T1-ENV-002 (replaces version 001: same lab and same people as Tasks 2–6)
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SIEM	Wazuh manager 4.9.2 (real) + Wazuh agent 4.9.2 deployed on the Parrot OS host
Parrot OS host	parrot-vm: a virtual machine (Debian 12, kernel 6.1.0-54-cloud-amd64, Parrot stand-in) under the QEMU hypervisor, built from a checksum-verified golden image (`767741ed83492b70`…)
macOS host	DAEDMAC1009 (the analyst's Mac): install.log and an SSH client trail shipped to Wazuh
People	The JML users from the Entra ID / Keycloak cycle: Sathya Hasi (joiner), Tara Jones (mover), Jeevan Smith (leaver), it_admin
Run	19:16:35–19:23:00 UTC, 2026-11-03

Summary

This task builds the single lab that all six Incident Response tasks use. The same machines, people and incident numbers run through to recovery in Task 6. A Wazuh manager runs on a management network. The Parrot OS host is a real virtual machine with the **Wazuh agent deployed and enrolled** (agent 001, status Active). The agent collects the host's authentication, kernel, EHR-audit and auditd logs and watches persistence paths in real time. macOS logs from the analyst's Mac are collected too: 500 of 500 sample lines received, none lost. Three custom rules turn the JML register into detections: a leaver logging in, a mover reading patient records through an old role, and root used by a role with no root entitlement. Each rule was tested with a line that must fire it and a near-miss that must not: **6/6 tests pass**. Wireshark has seven saved capture filters, all compiled. Volatility 3 has a symbol table built for this exact kernel and walked 82 processes in a baseline memory image. The clean state is saved as snapshot *baseline-clean*.

1. How this report meets the Task 1 requirements

Requirement	Where	Evidence
Wazuh installed and configured, Parrot OS in a hypervisor	§2, §3	transcripts/01, 03
Basic agent deployment	§3.2	transcripts/02 (agent_control, enrollment)
Log collection configuration from Parrot OS	§3.2, §4	lab/wazuh/agent.conf, transcripts/04
3 custom alert rules for security events	§5	lab/wazuh/can_ir1_core_rules.xml, wazuh/rule_tests.tsv

Requirement	Where	Evidence
Wireshark on Parrot OS with proper capture filters	§6	wireshark/cfilters, transcripts/05, t1_agent_traffic.pcapng
Volatility in Parrot OS with memory-analysis configuration	§7	transcripts/06, volatility/baseline_pslist.txt
Parrot OS and macOS logs ingested into Wazuh	§4	transcripts/04 (sent = received)
Documentation and evidence of functionality	all; App. A	EVIDENCE-INDEX-SHA256.txt

2. The lab

Everything runs in Docker Desktop with no administrator rights. VirtualBox cannot run here: it needs admin rights and does not run x86 guests on Apple silicon. So the Parrot OS host is a real VM under QEMU, inside a hypervisor container, and each hypervisor action maps to a VBoxManage command. The same lab carries the incident through Tasks 2–6.

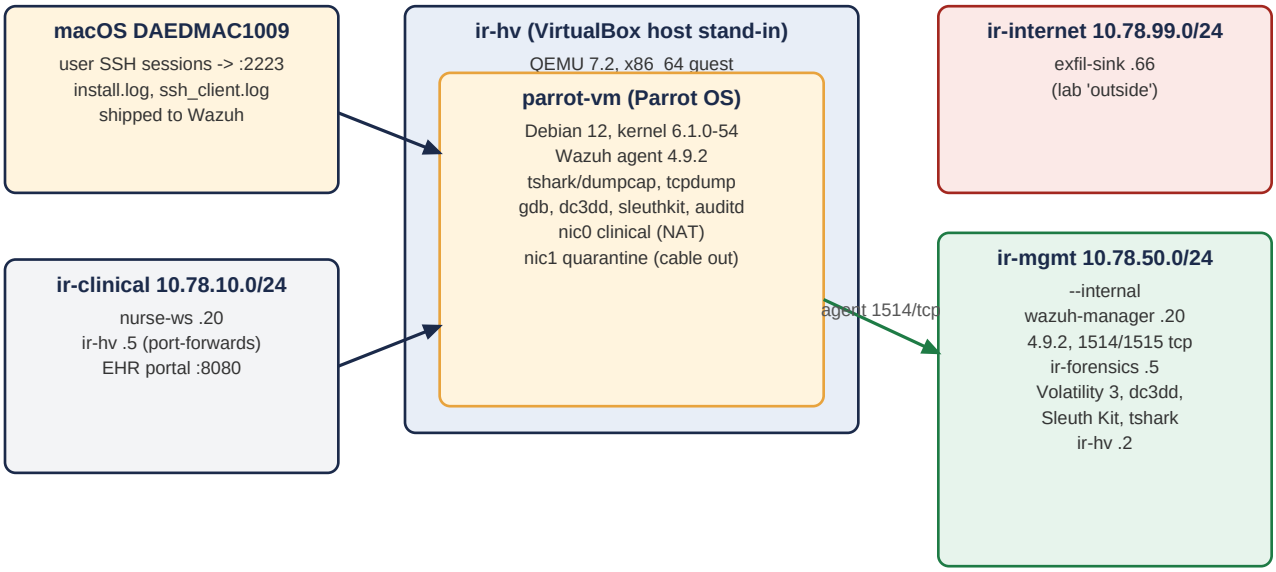

Figure 1. The IR1 lab (shared by Tasks 1–6).

Person (JML register)	Account on parrot-vm	JML event	State built in Task 1
Sathya Hasi	test.clinician	Joiner REQ-2026-0101 → Clinician	group clinical
Tara Jones	test.nurse	Mover CHG-2026-0201 Nurse → Billing	billing added; **clinical not removed** (as in the live lab)
Jeevan Smith	test.billing	Leaver TERM-2026-0301	locked, expired, DenyGroups; **SSH key still present**
IT administrator	it_admin	—	sudo

Person (JML register)	Account on parrot-vm	JML event	State built in Task 1
IR analyst	ir-analyst	—	forensic access (key only)

> **Scenario:** The two weaknesses in bold are deliberate. They mirror what the JML evidence showed in the live Entra ID / Keycloak lab, and they are what Tasks 2–6 detect, contain and fix.

Lab component	VirtualBox equivalent
QEMU VM from the golden image (checksum-verified Debian cloud image + IR toolset)	Parrot OS VM imported from an OVA
nic0: NAT with port forwards 22, 8080	Adapter 1: NAT + port forwarding
nic1: restricted network, link down	Adapter 2: internal network 'quarantine', cable disconnected
QMP savevm / dump-guest-memory / set_link	VBoxManage snapshot / debugvm dumpvmcore / setlinkstate

3. Wazuh

3.1 Manager

Manager version, content installed (transcripts/01, 03)

```
[2026-11-03T19:16:35Z] >>> lab_build
# IR1 shared lab build - 2026-11-03T19:16:35Z - host DAEDMAC1009 (macOS 26.3.1, arm64), Docker 29.7.2
$ docker network create ... (3 segments)
85f698d2d79493e9c268554d268c7406482e074f8d35d02ba58e04a019453923
can_ir1_core_rules.xml
local_rules.xml
4
wazuh-logcollector is running...
wazuh-remoted is running...
wazuh-analysisd is running...
wazuh-authd is running...

$ cat etc/lists/jml-leavers etc/lists/jml-movers etc/lists/rbac-no-root
== jml-leavers
test.billing:TERM-2026-0301
== jml-movers
test.nurse:billing
== rbac-no-root
```

3.2 Agent deployment on Parrot OS

The golden image installs `wazuh-agent 4.9.2` from Wazuh's signed repository. Task 1 writes the agent configuration (`lab/wazuh/agent.conf`): manager 10.78.50.20:1514/tcp, auto-enrollment, five log inputs, and real-time FIM on the home, cron, sudoers and sshd paths. It then starts the agent. The agent enrolled itself on 1515/tcp and connected.

Package (golden image)	Version
auditd	1:3.0.9-1
dc3dd	7.2.646-6
gdb	13.1-3
lsof	4.95.0-1
qemu-guest-agent	1:7.2+dfsg-7+deb12u18+b3
rsyslog	8.2302.0-1+deb12u1
sleuthkit	4.11.1+dfsg-1+b1
tcpdump	4.99.3-1
tshark	4.0.17-0+deb12u3
wazuh-agent	4.9.2-1

Manager side: agent 001 parrot-vm Active (transcripts/02)

```
$ agent_control -l   (manager side)

Wazuh agent_control. List of available agents:
   ID: 000, Name: wazuh-manager (server), IP: 127.0.0.1, Active/Local
   ID: 001, Name: parrot-vm, IP: any, Active

List of agentless devices:

$ agent_control -i 001
   Agent ID:   001
```

lab/wazuh/agent.conf - what the agent collects and watches

```
      <address>10.78.50.20</address>
  <localfile><log_format>syslog</log_format><location>/var/log/auth.log</location></localfile>
  <localfile><log_format>syslog</log_format><location>/var/log/ehr-audit.log</location></localfile>
  <localfile><log_format>syslog</log_format><location>/var/log/kern.log</location></localfile>
  <localfile><log_format>syslog</log_format><location>/var/log/ir1-test.log</location></localfile>
  <localfile><log_format>audit</log_format><location>/var/log/audit/audit.log</location></localfile>
    <directories realtime="yes">/home</directories>
    <directories realtime="yes">/var/spool/cron/crontabs</directories>
    <directories realtime="yes">/etc/sudoers.d</directories>
    <directories realtime="yes">/etc/ssh/sshd_config.d</directories>
```

4. Log collection: Parrot OS and macOS

Parrot OS. A tagged test line written on parrot-vm reached the manager through the agent (received 2026-11-03T19:21:52.838+0000, `parrot-vm(001)`, location `/var/log/ir1-test.log`). The agent reports which files it is reading:

Agent inputs (transcripts/04)

```
$ agent log inputs (agent side)
Analyzing file: '/var/log/audit/audit.log'.
Analyzing file: '/var/log/auth.log'.
Analyzing file: '/var/log/ehr-audit.log'.
Analyzing file: '/var/log/ir1-test.log'.
Analyzing file: '/var/log/kern.log'.
```

macOS. The Mac's unified log needs admin rights on this managed Mac (shown below), so the readable `/var/log/install.log` (51,823 lines) was normalised to ISO 8601. The newest 500 lines were shipped to the manager's macOS input to prove the path: sent 500, received 500, COMPLETE. Task 2 ships the whole file and the Mac's SSH client trail for analysis.

macOS host (transcripts/04)

```
$ sw_vers; hostname -s; date +%z
ProductName:        macOS
ProductVersion:     26.3.1
BuildVersion:       25D2128
DAEDMAC1009
-0500
$ log show --last 5m --predicate 'process == "sshd"'   (unified log needs admin on this managed Mac)
log: Could not open local log store: Operation not permitted
```

Events received by the manager, per source

```
$ events received per source (archives.json)
    500 "location":"/var/ossec/logs/macos/install.log"
      6 "location":"df -P"
      2 "location":"wazuh-agent"
      1 "location":"wazuh-remoted"
      1 "location":"wazuh-monitord"
      1 "location":"/var/log/ir1-test.log"
```

Reliability: Under emulation the manager's log collector can exit (seen once in testing). Shipping goes through `ship_file`: after every batch it checks the collector, restarts it and resends the batch if needed, and ends with a sent-versus-received count.

5. Three custom alert rules

Each rule ties a bundled Wazuh detection to the JML register through CDB lists, so the SIEM knows *who* should not be doing *what*. They are the rules that later detect the three incidents.

Rule	Level	Fires when	Incident it will catch
100210	10	sshd denies an account on the jml-leavers list (child of 5718/5710/5716/5760)	IR1-2026-001 Jeevan Smith
100221	12	EHR audit: a jml-movers account now in Billing views a record through group clinical (child of 100220)	IR1-2026-002 Tara Jones
100231	12	sudo to root by a role on rbac-no-root (child of 5402/5403)	IR1-2026-003 Sathya Hasi

lab/wazuh/can_ir1_core_rules.xml

```
<group name="care_access_navigator,ir1,core,">

  <!-- R1 leaver: a deprovisioned account tried to log in (-> IR1-2026-001). 5718 = sshd denied user.
    -->
  <rule id="100210" level="10">
    <if_sid>5718, 5710, 5716, 5760</if_sid>
    <list field="user" lookup="match_key">etc/lists/jml-leavers</list>
    <description>JML leaver account $(dstuser) attempted SSH login from $(srcip) - account was
    deprovisioned</description>
    <mitre>
      <id>T1078</id>
    </mitre>
    <group>authentication_failed,jml_leaver,hipaa_164.308.a.3.ii.c,</group>
  </rule>

  <!-- R2 mover: a mover opened a clinical record through access their new role does not grant (->
    IR1-2026-002). -->
  <rule id="100220" level="3">
    <decoded_as>ehr-audit</decoded_as>
    <description>EHR audit: $(dstuser) $(action) $(ehr_patient) - $(ehr_decision) via
    $(ehr_basis)</description>
    <group>ehr_access,</group>
  </rule>

  <rule id="100221" level="12">
    <if_sid>100220</if_sid>
    <field name="ehr_decision">^ALLOW$</field>
    <field name="ehr_basis">^group:clinical$</field>
    <list field="user" lookup="match_key_value" check_value="^billing$">etc/lists/jml-movers</list>
    <description>ePHI viewed by mover $(dstuser) (now Billing) through stale clinical group - record
    $(ehr_patient)</description>
    <mitre>
      <id>T1078</id>
    </mitre>
    <group>ehr_access,privilege_creep,hipaa_164.308.a.4,hipaa_164.312.a.1,</group>
  </rule>

  <!-- R3 joiner/RBAC: root obtained by a role with no root entitlement (-> IR1-2026-003). -->
  <rule id="100231" level="12">
    <if_sid>5402, 5403</if_sid>
    <!-- mirrors etc/lists/rbac-no-root (see note on rule 100222) -->
    <regex type="pcre2">^\s*(test\.clinician|test\.nurse|test\.billing)\s:</regex>
    <description>Root obtained via sudo by $(srcuser), whose role has no root entitlement (RBAC/SoD
    violation)</description>
    <mitre>
      <id>T1548.003</id>
    </mitre>
    <group>privilege_escalation,rbac_violation,hipaa_164.312.a.1,</group>
  </rule>

</group>
```

5.1 Rule tests

Each rule was replayed through `wazuh-logtest` with a line that must fire it and a near-miss that must not (expected '!id').

Rule	Test	Expected	Fired (level)	Result
100210	leaver test.billing denied by sshd	100210	100210 (10)	**PASS**
100210	near-miss: non-leaver denied	!100210	5718 (5)	**PASS**
100221	mover test.nurse views PT-1003	100221	100221 (12)	**PASS**
100221	near-miss: clinician views PT-1001	!100221	100220 (3)	**PASS**

Rule	Test	Expected	Fired (level)	Result
100231	test.clinician runs a command as root	100231	100231 (12)	**PASS**
100231	near-miss: it_admin (entitled) uses sudo	!100231	5403 (4)	**PASS**

wazuh-logtest -v trace for 100221 (transcripts/03)

```
$ wazuh-logtest -v (full trace for rule 100221)
**Phase 1: Completed pre-decoding.
**Phase 2: Completed decoding.
    dstuser: 'test.nurse'
    ehr_basis: 'group:clinical'
    ehr_decision: 'ALLOW'
    ehr_patient: 'PT-1003'
        *Rule 1 matched
    Trying rule: 89606 - Attempt to connect to screen sharing with username $(dstuser) from
    $(ip_address) failed.
    Trying rule: 89607 - Attempt to connect to screen sharing with username $(dstuser) from
    $(ip_address) succeeded.
    Trying rule: 100220 - EHR audit: $(dstuser) $(action) $(ehr_patient) - $(ehr_decision) via
    $(ehr_basis)
        *Rule 100220 matched
    Trying rule: 100221 - ePHI viewed by mover $(dstuser) (now Billing) through stale clinical group -
    record $(ehr_patie
        *Rule 100221 matched
**Phase 3: Completed filtering (rules).
    id: '100221'
    level: '12'
    description: 'ePHI viewed by mover test.nurse (now Billing) through stale clinical group - record
    PT-1003'
    groups: '['care_access_navigator', 'ir1', 'core', 'ehr_access', 'privilege_creep']'
    mitre.id: '['T1078']'
**Alert to be generated.
```

6. Wireshark on Parrot OS

Seven named capture filters were saved to `~/.config/wireshark/cfilters` for the analyst. Wireshark's GUI and tshark both read this file. Each filter was compiled to BPF on the clinical interface with `dumpcap -d`, so a typo fails now, not mid-incident. One filter was then used for a live capture.

Saved filter	BPF instructions
tcp port 22	20
tcp port 8080 and host 10.0.2.15	18
tcp port 1514 or tcp port 1515	23
host 10.78.99.66	14
src host 10.0.2.15 and not port 22 and not port 1514 and not port 1515	25
udp port 53	20
tcp[tcpflags] & (tcp-syn|tcp-ack) == tcp-syn	12

Live capture with 'IR1 Wazuh agent traffic' (transcripts/05)

```
# live capture with a saved filter: 20 s of Wazuh agent traffic
================================================================================
TCP Conversations
Filter:<No Filter>
                                                      |       <-      | |       ->      | |
    Total     |    Re
                                                      | Frames  Bytes | | Frames  Bytes | | Frames
      Bytes |
10.0.2.15:50688           <-> 10.78.50.20:1514                4 305 bytes       4 1202 bytes       8
    1507 bytes
================================================================================
Number of packets:   8
acquired /dev/shm/t1_agent.pcapng ->
    /ir/task1_environment_setup/evidence/wireshark/t1_agent_traffic.pcapng  bytes=217
  sha256 source(parrot-vm)=8187ac9205885bb6957e8d9d67b39d0217bc6c4b0762c7f9a2650991ed1df34a
  sha256 copy(ir-forensics)=8187ac9205885bb6957e8d9d67b39d0217bc6c4b0762c7f9a2650991ed1df34a  MATCH
```

7. Volatility 3

Volatility 3 runs on the Parrot forensic workstation. It needs a symbol table (ISF) that matches the exact kernel in the image. It was built with the Volatility Foundation's `dwarf2json` v0.9.0 from Debian's debug kernel package. To prove the configuration works, the hypervisor dumped the clean VM's RAM and Volatility walked its process list: 82 processes, including the Wazuh agent.

transcripts/06_volatility.txt

```
$ vol -h | head -1; uname -r on parrot-vm
Volatility 3 Framework 2.7.0
6.1.0-54-cloud-amd64
$ ls -l /opt/symbols/linux; sha256sum
total 1756
-rw-r--r-- 1 root root 1797916 Nov  2 18:37 debian-6.1.0-54-cloud-amd64.json.xz
c69b86fa8a29404ba410fa3fc3b3118b8e62ffb1b58b3fd161280cf4c3c0042f
    /opt/symbols/linux/debian-6.1.0-54-cloud-amd64.json.

# baseline RAM image of the clean host, from the hypervisor   [VBoxManage debugvm parrot-vm dumpvmcore]
{"return": {}}
{"return": {"total": 1074003968, "status": "completed", "completed": 1074003968}}
sha256 79c3b2f3c40dc3da3a779415a42f980d0170bdaf96501127a8346ebd3da941f1  parrot-vm_baseline.mem.elf

$ vol -s /opt/symbols -f parrot-vm_baseline.mem.elf banners.Banners
0xfb1fba0   Linux version 6.1.0-54-cloud-amd64 (debian-kernel@lists.debian.org) (gcc-12 (Debian
    12.2.0-14+deb12u1) 12.2.
0x10cf54c0  Linux version 6.1.0-54-cloud-amd64 (debian-kernel@lists.debian.org) (gcc-12 (Debian
    12.2.0-14+deb12u1) 12.2

$ vol ... linux.pslist.PsList   (config works if the kernel's process list is walked)
processes found: 82
OFFSET (V)  PID TID PPID    COMM    File output
0x8b17b7210000  28  28  2   kauditd Disabled
0x8b17b705a0c0  326 326 1   auditd  Disabled
0x8b17b667a0c0  811 811 1   sshd    Disabled
0x8b17b6678000  852 852 1   python3 Disabled
0x8b178620a0c0  1078    1078    1   wazuh-execd Disabled
0x8b1785d7a0c0  1089    1089    1   wazuh-agentd    Disabled
0x8b1784ff8000  1103    1103    1   wazuh-syscheckd Disabled
0x8b1785da0000  1131    1131    1   wazuh-logcollec Disabled
0x8b178622a0c0  1149    1149    1   wazuh-modulesd  Disabled
```

Configuration point: The guest CPU is configured `-cpu max,la57=off`. With 5-level paging on, Volatility 3 finds the kernel but no processes, which was found in an earlier run. VirtualBox guests normally use 4-level paging.

8. Baseline

Reachability of the clean environment, re-run in Tasks 3 and 6:

#	Path	State
P1	macOS host -> parrot-vm SSH (127.0.0.1:2223)	REACHABLE
P2	clinical LAN nurse-ws -> parrot-vm SSH (10.78.10.5:22)	REACHABLE
P3	clinical LAN nurse-ws -> EHR portal (10.78.10.5:8080)	REACHABLE
P4	parrot-vm -> exfil-sink (10.78.99.66:8080)	REACHABLE
P5	parrot-vm -> internet (deb.debian.org:80)	REACHABLE (HTTP 200)
P6	ir-forensics -> parrot-vm quarantine SSH (10.78.50.2:2222)	BLOCKED

Snapshot baseline-clean (used by Task 6)

```
# known-good snapshot used by Task 6   [VBoxManage snapshot parrot-vm take baseline-clean --live]
(ok)
List of snapshots present on all disks:
ID        TAG              VM SIZE                DATE     VM CLOCK     ICOUNT
--        baseline-clean    501 MiB 2026-11-03 19:22:59 00:06:18.221
```

9. Substitutions and limitations

- VirtualBox → QEMU, and Parrot OS → Debian 12 with the Parrot IR toolset. The commands are identical; confirm with the instructor.
- No Wazuh dashboard. The indexer and dashboard images exist only for x86 and would run under emulation beside the VM on an 8 GB Docker VM. Ingestion is shown from the manager itself (archives.json and alerts.json), which is the data the dashboard displays.
- macOS logs are shipped as files: a macOS agent needs admin rights on this managed Mac.

Appendix A — Evidence index

evidence/EVIDENCE-INDEX-SHA256.txt

```
f6bc16daae204bc8...  ./macos/install_iso.log
221483aaf4cbfb2c...  ./macos/install_sample500.log
79ae0bad9b0b2bff...  ./macos/install.log
47ba2ca8d56b14ea...  ./markers.csv
805ad65c97aff402...  ./transcripts/01_lab_build.txt
1c218ce1bedc110c...  ./transcripts/02_provision_agent.txt
cc3e363a40b98401...  ./transcripts/03_wazuh_rules.txt
0e228ab66bda9d31...  ./transcripts/04_log_collection.txt
5848ca5d29ee3204...  ./transcripts/05_wireshark.txt
3854721186972002...  ./transcripts/06_volatility.txt
33281966627006d1...  ./volatility/baseline_pslist.txt
be2d1bd69535027b...  ./wazuh/can_ir1_core_rules.xml
25af586ecbdfa08b...  ./wazuh/rule_tests.tsv
847a7f62ab0d2c07...  ./wireshark/cfilters
8187ac9205885bb6...  ./wireshark/t1_agent_traffic.pcapng
# outside git: /Users/Adult/ir1-vm-lab/evidence/baseline
79c3b2f3c40dc3da...  parrot-vm_baseline.mem.elf
```